Facet: Shared Core Architecture — Rust as Cross-Platform Foundation for Helix VPN

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Executive Summary

This research evaluates Rust as a shared core library foundation for maximum code reuse across all Helix VPN target platforms: macOS, Windows, Linux, Android, iOS, HarmonyOS, Aurora OS, and Web. The analysis covers FFI patterns, binding generators (UniFFI, flutter_rust_bridge, wasm-bindgen), build systems, real-world case studies from Mullvad VPN, Signal, Cloudflare BoringTun, and provides a recommended architecture with estimated code reuse percentages per platform.

Key Finding: A well-architected Rust shared core can achieve **75-85% code reuse** across desktop platforms, **60-70%** on mobile (Android/iOS), and **40-50%** on Web (WASM), with HarmonyOS/Aurora OS achieving **60-70%** through Linux-compatible targets.

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1. Rust Cross-Compilation Target Matrix

1.1 Supported Target Triples for All Platforms

Rust's LLVM-based compiler backend provides first-class cross-compilation support. The following target triples are relevant for Helix VPN [^342^]:

Platform	Target Triple	Tier	Status
macOS (Intel)	`x86_64-apple-darwin`	Tier 2	Production-ready
macOS (Apple Silicon)	`aarch64-apple-darwin`	Tier 2	Production-ready
iOS (Device)	`aarch64-apple-ios`	Tier 2	Production-ready
iOS (Simulator - Apple Silicon)	`aarch64-apple-ios-sim`	Tier 2	Production-ready
iOS (Simulator - Intel)	`x86_64-apple-ios`	Tier 2	Production-ready
Windows (x64)	`x86_64-pc-windows-msvc`	Tier 1	Full support
Windows (ARM64)	`aarch64-pc-windows-msvc`	Tier 2	Supported
Linux (x64)	`x86_64-unknown-linux-gnu`	Tier 1	Full support
Linux (ARM64)	`aarch64-unknown-linux-gnu`	Tier 1	Full support
Android (ARM64)	`aarch64-linux-android`	Tier 2	Production-ready
Android (ARMv7)	`armv7-linux-androideabi`	Tier 2	Production-ready
Android (x86_64)			

Platform	Target Triple	Tier	Status
	`x86_64-linux-android`	Tier 2	Production-ready
Web/Browser	`wasm32-unknown-unknown`	Tier 2	Production-ready
HarmonyOS	`aarch64-unknown-linux-ohos`	Tier 2	Supported with SDK setup

1.2 Cross-Compilation Tools

cross-rs (formerly cargo-cross)

The community-maintained wrapper around `cargo` that makes cross-compilation painless using Docker containers [^367^]:

```
# Install
cargo install cross --git https://github.com/cross-rs/cross

# Build for any target
cross build --release --target aarch64-linux-android
cross build --release --target aarch64-apple-ios
cross build --release --target x86_64-pc-windows-msvc
```

Key advantage: Provides complete cross-compilation environments with correct toolchains and system libraries, eliminating manual linker configuration [^367^].

rustup Target Management

```
# Add all required targets for mobile development
rustup target add aarch64-apple-ios aarch64-apple-ios-sim x86_64-
        apple-ios
rustup target add aarch64-linux-android armv7-linux-androideabi
        x86_64-linux-android
rustup target add x86_64-pc-windows-msvc aarch64-pc-windows-msvc
rustup target add wasm32-unknown-unknown
```

---

2. UniFFI: Mozilla's Multi-Language Binding Generator

2.1 Overview

UniFFI is a toolkit for building cross-platform software components in Rust, developed by Mozilla and used extensively in Firefox mobile and desktop browsers [^338^]. It automatically generates foreign-language bindings, eliminating the need to write FFI plumbing by hand.

> "Instead, uniffi does most of the work for us by generating the plumbing necessary to transport data across the FFI, including the specific language bindings, making it a little easier to write things once and a lot easier to maintain multiple supported languages." [^336^]

2.2 Supported Languages

Language	Support Level	Notes
Kotlin	Full (1st party)	Android primary target
Swift	Full (1st party)	iOS/macOS primary target
Python	Full (1st party)	Desktop scripting
Ruby	Legacy/partial	Community maintained
C#	3rd party	Community bindings available
Go	3rd party	Community bindings available
Dart	3rd party	Via flutter_rust_bridge
JavaScript	3rd party	WASM + React Native TurboModules

2.3 Interface Definition

UniFFI supports two interface definition approaches [^340^]:

UDL (UniFFI Definition Language) - WebIDL-based:

```
namespace helixvpn {
    string connect_to_server(string server_address);
    void disconnect();
    TunnelState get_tunnel_state();
};
```

```
interface TunnelConfig {
    constructor(string server_address, u16 port);
    string get_server_address();
};

enum TunnelState {
    "Disconnected",
    "Connecting",
    "Connected",
    "Error"
};
```

Proc-Macro Approach (modern, preferred):

```
#[uniffi::export]
pub fn connect_to_server(server_address: String) -> String {
    // implementation
}

#[derive(uniffi::Object)]
pub struct TunnelConfig {
    server_address: String,
    port: u16,
}

#[uniffi::export]
impl TunnelConfig {
    #[uniffi::constructor]
    pub fn new(server_address: String, port: u16) -> Self {
        Self { server_address, port }
    }
}
```

2.4 Maturity and Production Usage

- **Mozilla Firefox**: Used in Glean SDK (telemetry), Nimbus SDK (experimentation), and application-services [^336^] [^338^]
- **Production readiness**: "We consider it ready for production use" [^338^]
- **Version status**: Pre-1.0, but actively maintained with breaking changes minimized for simple consumers
- **Community ecosystem**: Third-party bindings for C#, Go, Dart, Java, Node.js [^338^]

2.5 Limitations

- Async support is evolving: "The futures/async support is quite immature" [^334^]
- Advanced features may break between upgrades [^338^]

- Some advanced FFI patterns (e.g., custom tokio runtime management) may require manual FFI [^357^]

2.6 Alternative: BoltFFI

A newer high-performance alternative claiming significant speed improvements [^341^]:

- `echo_i32`: <1 ns vs 1,416 ns (UniFFI) — **>1000x faster**
- Supports Swift, Kotlin, TypeScript (WASM) with Python planned
- Handles XCFramework generation for Apple platforms natively

---

3. FFI Patterns and Best Practices

3.1 Core Principles

1. **Minimize `unsafe` code**: Isolate unsafe to thin FFI boundary layers
2. **C ABI compatibility**: Use `#[no_mangle] pub extern "C"` for all exported functions
3. **Memory safety**: Define clear ownership semantics across the boundary
4. **Error handling**: Convert Rust `Result` to language-appropriate error types

3.2 Async Rust Across FFI

The `async-ffi` crate provides FFI-compatible Futures [^330^]:

```rust
use async_ffi::{FfiFuture, FutureExt};

#[no_mangle]
pub extern "C" fn work(arg: u32) -> FfiFuture<u32> {
    async move {
        let ret = do_some_io(arg).await;
        ret
    }.into_ffi()
}
```

Key considerations [^329^] [^333^]:

- Use a static/shared Tokio Runtime rather than creating one per call
- Thread-local storage won't work correctly across dynamic library boundaries
- Pass runtime handles explicitly via FFI if needed

Tokio Runtime pattern for FFI [^329^]:

```
use std::sync::LazyLock;
use tokio::runtime::Runtime;

static RUNTIME: LazyLock<Runtime> = LazyLock::new(|| {
    Runtime::new_multi_thread()
        .worker_threads(4)
        .enable_all()
        .build()
        .expect("Failed to create Tokio runtime")
});

#[no_mangle]
pub extern "C" fn vpn_connect(config: *const c_char) -> i32 {
    let config_str = unsafe {
        CStr::from_ptr(config).to_string_lossy() };
    match RUNTIME.block_on(async_connect(config_str.as_ref())) {
        Ok(_) => 0,
        Err(e) => e.code(),
    }
}
```

3.3 Error Handling Across Boundaries

Pattern 1: Error codes with thread-local error messages [^331^]:

```
thread_local! { static LAST_ERROR: RefCell<Option<String>> =
        RefCell::new(None); }

#[no_mangle]
pub extern "C" fn get_last_error() -> *mut c_char {
    LAST_ERROR.with(|e| {
        e.borrow().as_ref()
            .map(|s| CString::new(s.as_str()).unwrap().into_raw())
            .unwrap_or(std::ptr::null_mut())
    })
}
```

Pattern 2: Return structs with error info:

```
#[repr(C)]
pub struct FfiResult<T> {
    pub data: T,
    pub error_code: i32,
    pub error_message: *mut c_char,
}
```

Pattern 3: UniFFI-style (recommended): UniFFI automatically converts Rust `Result<T, E>` to exceptions/errors in target languages [^340^].

3.4 Memory Management Best Practices

- **String passing**: Rust returns `CString` → foreign language frees via dedicated function
- **Object handles**: Use opaque pointer types (`*mut c_void`) with create/destroy functions
- **Avoid panics across FFI**: Use `catch_unwind` at the boundary (undefined behavior otherwise)
- **Zero-copy where possible**: Use `&[u8]` slices for packet data transfer

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4. Android Rust Integration

4.1 Toolchain: cargo-ndk

`cargo-ndk` is the standard tool for building Rust for Android [^348^]:

```
# Install
cargo install cargo-ndk

# Add Android targets
rustup target add aarch64-linux-android armv7-linux-androideabi
        x86_64-linux-android

# Build for all architectures
cargo ndk --target aarch64-linux-android --android-platform 33 --
        build --release
cargo ndk --target armv7-linux-androideabi --android-platform 33
        -- build --release
cargo ndk --target x86_64-linux-android --android-platform 33 --
        build --release
```

Key features [^348^]:

- Automatically finds and sets NDK linker and AR tools
- Handles sysroot configuration
- Supports API level specification

4.2 JNI Bindings: jni-rs

The `jni` crate (version 0.21+) provides Rust JNI bindings [^344^]:

```
use jni::JNIEnv;
use jni::objects::JClass;
```

```rust
use jni::signature::JavaType;

#[no_mangle]
pub extern "C" fn Java_com_helixvpn_core_RustBridge_connect(
    mut env: JNIEnv,
    _class: JClass,
    server_address: JString,
) -> jint {
    let addr: String = env.get_string(&server_address)
        .expect("Invalid string")
        .into();

    match vpn_connect(&addr) {
        Ok(_) => 0,
        Err(e) => {
            env.throw_new("com/helixvpn/core/VpnException",
        e.to_string())
                .unwrap();
            -1
        }
    }
}
```

4.3 Gradle Plugin Integration

The Cargo NDK Gradle Plugin automates Rust builds within Android projects [^346^]:

```
// root build.gradle
buildscript {
    dependencies {
        classpath
        "gradle.plugin.com.github.willir.rust:plugin:0.3.4"
    }
}

// app/build.gradle
apply plugin: "com.github.willir.rust.cargo-ndk-android"

cargoNdk {
    targets = ["arm64", "arm", "x86_64"]
    module = "../rust-core"
    apiLevel = 24
    buildTypes {
        release { buildType = "release" }
        debug { buildType = "debug" }
    }
}
```

This plugin [^346^]:

- Runs automatically during `./gradlew assembleDebug` / `assembleRelease`
- Copies `.so` files to `jniLibs/` directory
- Supports per-build-type configuration
- Allows target filtering via gradle properties (`-Prust-target=arm64`)

4.4 Android NDK Rust Toolchain Support

- NDK r23+ provides improved LLVM toolchain compatibility
- Rust Android targets use the NDK's Clang linker
- Minimum supported API level: 24 (Android 7.0) for full Rust std support [^344^]

4.5 .so Library Packaging

Output structure:

```
android/app/src/main/jniLibs/
├── arm64-v8a/libhelix_core.so       (from aarch64-linux-android)
├── armeabi-v7a/libhelix_core.so     (from armv7-linux-
androideabi)
├── x86_64/libhelix_core.so          (from x86_64-linux-android)
```

---

5. iOS Rust Integration

5.1 Modern Approach: XCFramework + UniFFI

The modern recommended approach uses `xcodebuild -create-xcframework` with UniFFI-generated bindings, replacing the deprecated `cargo-lipo` tool [^349^]:

```
# 1. Add iOS targets
rustup target add aarch64-apple-ios aarch64-apple-ios-sim

# 2. Build for device
cargo build --release --target=aarch64-apple-ios

# 3. Build for simulator
cargo build --release --target=aarch64-apple-ios-sim

# 4. Generate UniFFI bindings
cargo run --bin uniffi-bindgen generate \
  --library ./target/aarch64-apple-ios/release/libhelix.dylib \
  --language swift --out-dir ./bindings
```

```
# 5. Rename modulemap (critical step!)
mv bindings/helixFFI.modulemap bindings/module.modulemap

# 6. Create XCFramework
xcodebuild -create-xcframework \
  -library ./target/aarch64-apple-ios-sim/release/libhelix.a
        -headers ./bindings \
  -library ./target/aarch64-apple-ios/release/libhelix.a
        -headers ./bindings \
  -output ios/HelixCore.xcframework
```

5.2 Swift Integration

After importing the XCFramework and generated Swift file [^82^]:

```
import SwiftUI
import helixFFI  // The generated UniFFI module

struct ContentView: View {
    @State private var status = "Disconnected"

    func connect() {
        // Call Rust function through UniFFI-generated bindings
        let result = connectToServer(serverAddress: "us-
        east.helix.vpn")
        status = "Connected"
    }
}
```

5.3 Legacy: cargo-lipo (Deprecated)

`cargo-lipo` was previously used to create universal fat binaries but is now deprecated because `arm64` can represent both device and simulator targets [^349^]. Use XCFrameworks instead.

5.4 cbindgen for C Headers

When not using UniFFI, `cbindgen` generates C headers from Rust code:

```
cargo install cbindgencbindgen --output include/helix_core.h
```

Used by Signal's libsignal for manual bridge layers [^362^].

5.5 Key Considerations for iOS

- **XCFramework is required** (not optional) for Apple Silicon Macs since the same architecture (arm64) is used for both devices and simulators [^349^]

- **Module map renaming**: Must rename `xxxFFI.modulemap` to `module.modulemap` for Xcode detection [^82^]
- **libresolv.tbd**: Must be linked on iOS for DNS resolution [^345^]
- **Bitcode**: Disabled by default in modern Xcode; Rust binaries are not bitcode-compatible

---

6. Desktop Rust Integration

6.1 Tauri Commands

Tauri provides a Rust backend with web frontend, communicating via typed commands [^10^]:

```
// src-tauri/src/main.rs
#[tauri::command]
async fn vpn_connect(state: tauri::State<'_, VpnState>, server:
        String) -> Result<String, String> {
    state.core.connect(&server).await
        .map(|_| "Connected".into())
        .map_err(|e| e.to_string())
}

fn main() {
    tauri::Builder::default()
        .manage(VpnState::default())
        .invoke_handler(tauri::generate_handler![vpn_connect,
        vpn_disconnect])
        .run(tauri::generate_context!())
        .expect("error while running tauri application");
}
```

Frontend (TypeScript):

```
import { invoke } from '@tauri-apps/api/core';

async function connect() {
    const result = await invoke('vpn_connect', { server: 'us-
        east.helix.vpn' });
    console.log(result);
}
```

Tauri advantages [^10^]:

- Tiny bundle sizes (2-10 MB)
- Native memory efficiency
- Full OS API access through Rust
- Desktop is production-stable; mobile support in beta

6.2 Flutter Desktop with Rust

Two approaches for Flutter + Rust on desktop:

Approach A: flutter_rust_bridge (Recommended)

The `flutter_rust_bridge` crate is a Flutter Favorite and generates high-level bindings [^392^]:

```
# One-liner setup
cargo install flutter_rust_bridge_codegen &&
        flutter_rust_bridge_codegen create my_app
```

Approach B: Direct FFI via dart:ffi [^379^]:

```
// Load the Rust library
final lib = Platform.isAndroid
    ? DynamicLibrary.open('libhelix_core.so')
    : Platform.isIOS
        ? DynamicLibrary.process()  // Statically linked
        : DynamicLibrary.open('libhelix_core.dylib');  // macOS

final connect = lib.lookupFunction<ConnectNative,
        ConnectDart>('rust_connect');
```

Desktop library locations:

- **macOS**: `HelixVPN.app/Contents/Frameworks/libhelix_core.dylib`
- **Windows**: Place `.dll` next to executable or in `PATH`
- **Linux**: Standard library path or `LD_LIBRARY_PATH`

6.3 Direct Dynamic Library Loading

For native desktop apps (non-Flutter, non-Tauri):

```
// Loading Rust core as .so/.dll/.dylib
#[cfg(target_os = "macos")]
const LIB_PATH: &str = "libhelix_core.dylib";
#[cfg(target_os = "linux")]
const LIB_PATH: &str = "libhelix_core.so";
#[cfg(target_os = "windows")]
const LIB_PATH: &str = "helix_core.dll";

let lib = unsafe { libloading::Library::new(LIB_PATH)? };
let connect: Symbol<unsafe extern "C" fn(*const c_char) -> i32> =
    unsafe { lib.get(b"vpn_connect")? };
```

---

7. WASM Target for Browser/Extensions

7.1 wasm-bindgen + wasm-pack

The standard Rust-to-WASM toolchain [^353^] [^354^]:

```
# Install
rustup target add wasm32-unknown-unknown
cargo install wasm-pack

# Build for browser
wasm-pack build --target web --out-dir pkg/

# Build for bundler (webpack/rollup)
wasm-pack build --target bundler --out-dir pkg/
```

7.2 VPN Core in WebAssembly

For a browser extension or web UI, the Rust VPN core compiles to WASM [^354^]:

```
// src/lib.rs
use wasm_bindgen::prelude::*;

#[wasm_bindgen]
pub struct WasmVpnCore {
    inner: VpnCore,
}

#[wasm_bindgen]
impl WasmVpnCore {
    #[wasm_bindgen(constructor)]
    pub fn new() -> Self {
        console_error_panic_hook::set_once();
        Self { inner: VpnCore::new() }
    }

    pub async fn connect(&self, server: String) ->
        Result<JsValue, JsValue> {
        self.inner.connect(&server).await
            .map(|state| JsValue::from_str(&format!("{:?}",
        state)))
            .map_err(|e| JsValue::from_str(&e.to_string()))
    }
}
```

7.3 Web Limitations for VPN

Critical: WASM in browsers **cannot** create raw network sockets or TUN interfaces due to browser sandboxing. For a browser extension VPN:

1. **Use proxy-based approach**: Configure browser proxy settings to route through a VPN gateway
2. **WebRTC for P2P**: Use WebRTC data channels for VPN tunneling
3. **Native messaging**: Communicate with a native host application that runs the actual VPN core
4. **WASM for crypto**: Use Rust crypto (Noise protocol, ChaCha20-Poly1305) in WASM while delegating networking to JavaScript APIs

7.4 Browser Extension Architecture

```
Browser Extension Architecture:

┌─────────────────────────────────────────────────┐
│  Popup UI (React/Vue)                           │
│  └─ Calls WASM crypto functions                 │
├─────────────────────────────────────────────────┤
│  Background Service Worker                      │
│  └─ chrome.proxy API or Native Messaging        │
├─────────────────────────────────────────────────┤
│  Native Host App (Rust binary, platform-specific)   │
│  └─ Real TUN interface, WireGuard, etc.         │
└─────────────────────────────────────────────────┘
```

7.5 Performance Considerations

- WASM has ~1.5-2x overhead vs native Rust [^353^]
- Crypto operations in WASM are still faster than pure JavaScript
- Use `wasm-opt` (Binaryen) for additional optimization
- Enable `wee_alloc` or `dlmalloc` as smaller WASM allocators

---

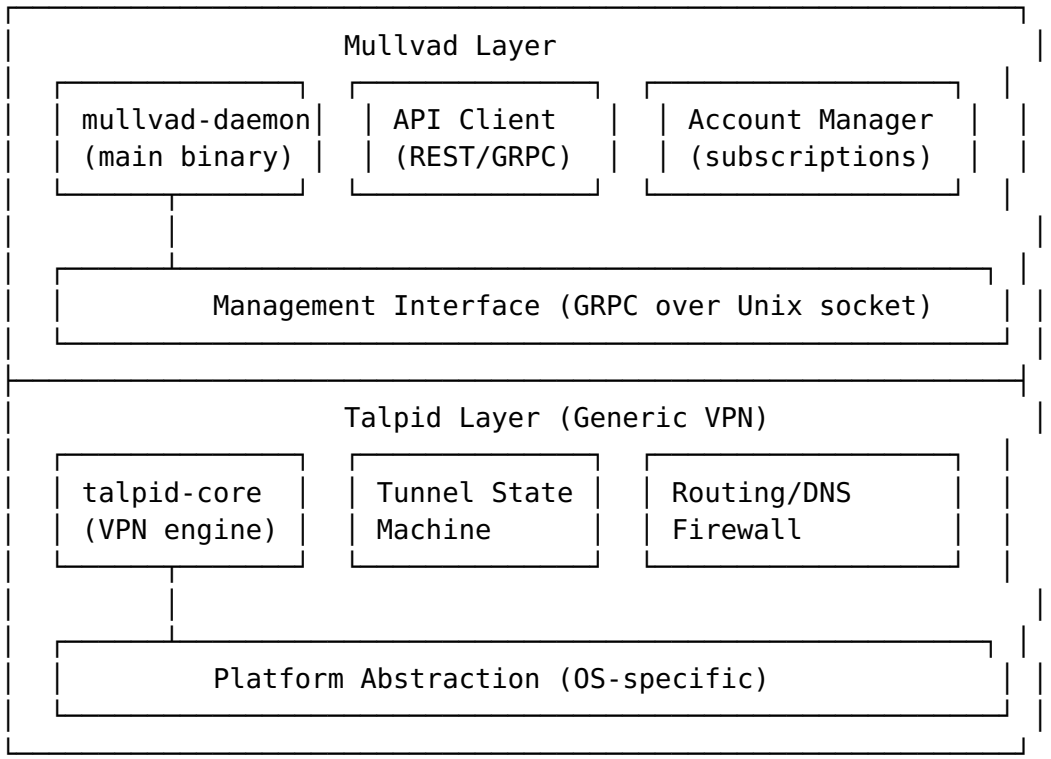
8. Real-World Case Studies

8.1 Mullvad VPN — Rust Core Architecture

Repository: `github.com/mullvad/mullvadvpn-app` (7.3k+ stars) [^155^]

Architecture Overview

Mullvad's architecture splits into two distinct layers [^23^]:



Key Design Principles

> "The `talpid` crates are supposed to be completely unrelated to Mullvad specific things. A `talpid` crate is not allowed to know anything about the API through which the daemon fetch Mullvad account details or download VPN server lists." [^23^]

- **talpid-core**: Generic VPN client library, completely Mullvad-agnostic [^155^]
- **mullvad-daemon**: Mullvad-specific daemon binary
- **Frontend communication**: GRPC over Unix socket (desktop), JNI (Android), standalone iOS

Platform Split

Platform	Architecture	Code Reuse
Desktop (macOS/ Windows/Linux)	Shared `mullvad-daemon` Rust binary + Electron GUI	~85%
Android	Same Rust core via JNI + Kotlin UI	~70%
iOS		Separate codebase

Platform	Architecture	Code Reuse
	Standalone Swift implementation in `ios/` directory	

GotaTun — WireGuard in Rust

Mullvad forked Cloudflare's BoringTun to create GotaTun [^363^] [^61^]:

> "GotaTun integrates additional functionality like DAITA and Multihop compared to Cloudflare's BoringTun code... Previously Mullvad was relying on a Go language implementation of WireGuard. With the WireGuard Go implementation they had encountered crashes while so far 'not a single crash' has been detected with GotaTun." [^61^]

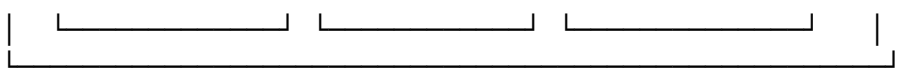
- Forked from BoringTun (BSD-3-clause → MPL-2.0)
- Adds first-class Android support, DAITA, Multihop
- Replaced WireGuard-Go across all platforms in 2026

8.2 Signal — libsignal Rust Core

Repository: `github.com/signalapp/libsignal` [^362^]

Architecture

```
libsignal Architecture:
┌──────────────────────────────────────────────────────┐
│  Client Apps (Android/iOS/Desktop)                   │
├──────────┬──────────┬────────────────────────────────┤
│  Java    │  Swift   │  TypeScript (Node/Electron)    │
│  (JNI)   │  (C FFI) │  (N-API)                       │
├──────────┴──────────┴────────────────────────────────┤
│  Bridge Layer (language-specific wrappers)           │
│  java/ │ swift/ │ node/                              │
├──────────────────────────────────────────────────────┤
│  Rust Core Crates                                    │
│  ┌──────────────┐ ┌──────────────┐ ┌──────────────┐  │
│  │ libsignal-   │ │ signal-      │ │ device-      │  │
│  │ protocol     │ │ crypto       │ │ transfer     │  │
│  │ (Double      │ │ (AES-GCM,    │ │ (QR codes)   │  │
│  │  Ratchet)    │ │  primitives)│ │              │  │
│  └──────────────┘ └──────────────┘ └──────────────┘  │
│  ┌──────────────┐ ┌──────────────┐ ┌──────────────┐  │
│  │ zkgroup      │ │ account-     │ │ usernames    │  │
│  │ (ZK proofs)  │ │ keys         │ │              │  │
```



Key insight: Signal uses **manual FFI bindings** (not UniFFI) [^357^]:

> "Unlike Proton that we studied previously, Signal doesn't use Mozilla's UniFFI to automatically generate Kotlin and Swift bindings and instead rely on manual bindings using FFI (Foreign Function Interface) and JNI (Java Native Interface) wrappers."

Why manual FFI? [^357^]

- Already had wrapping code from previous `libsignal-protocol-c` library
- Need advanced control over tokio runtime across FFI boundaries
- UniFFI's async handling may be incompatible with their runtime requirements

Platform support matrix [^360^]:

Platform	Architectures	Interface
Android	arm64, armv7, x64	JNI
iOS	arm64	C FFI + Swift
macOS	arm64, x64	Node N-API
Linux	arm64, x64	Node N-API
Windows	x64	Node N-API
Web	WASM	wasm-bindgen

8.3 Cloudflare BoringTun

Repository: `github.com/cloudflare/boringtun` [^49^]

> "BoringTun is an implementation of the WireGuard protocol designed for portability and speed... successfully deployed on millions of iOS and Android consumer devices as well as thousands of Cloudflare Linux servers." [^49^]

Architecture [^59^]:

- Library (`boringtun`): Core WireGuard protocol without network/tunnel stacks
- CLI (`boringtun-cli`): Userspace WireGuard for Linux/macOS
- Platform-idiomatic tunnel stacks implemented separately

Supported platforms [^49^]:

Target	Binary	Library
x86_64-unknown-linux-gnu	Yes	Yes
aarch64-unknown-linux-gnu	Yes	Yes
x86_64-apple-darwin	Yes	Yes
x86_64-pc-windows-msvc	No	Yes
aarch64-apple-ios	No	Yes
aarch64-linux-android	No	Yes
arm-linux-androideabi	No	Yes

Key insight from Cloudflare [^59^]:

> "After we decided to create a userspace WireGuard implementation, there was the small matter of choosing the right language... The obvious answer was Rust. Rust is a modern, safe language that is both as fast as C++ and is arguably safer than Go."

8.4 Comparison Summary

Project	Rust Core %	Binding Strategy	Platforms
Mullvad VPN	~85% desktop, ~70% Android	Manual FFI + JNI + GRPC	macOS, Win, Linux, Android, iOS
Signal	~80%	Manual FFI + JNI + N-API	Android, iOS, Desktop, Web
BoringTun	~95% (library)	C ABI + JNI	Linux, macOS, iOS, Android, Windows
GotaTun	~95%	C ABI + JNI	All Mullvad platforms

---

9. VPN-Specific Rust Crates

9.1 TUN/TAP Interface: tun-rs

Crate: `tun-rs` — Cross-platform TUN/TAP library [^364^]

```rust
use tun_rs::DeviceBuilder;

// Async TUN with Tokio
let dev = DeviceBuilder::new()
    .name("utun7")
    .ipv4("10.0.0.1", 24, None)
```

```
    .mtu(1400)
    .build_async()?;  // Returns AsyncDevice

// Read packets asynchronously
let mut buf = vec![0; 65536];
loop {
    let len = dev.recv(&mut buf).await?;
    process_packet(&buf[..len]).await;
}
```

Platform support: Windows, Linux, macOS, FreeBSD, OpenBSD, NetBSD, Android, iOS, tvOS, and **OpenHarmony** [^364^]

Mobile integration:

```
// iOS: Use file descriptor from PacketTunnelProvider
// Android: Use fd from VpnService.Builder.establish()
let fd = obtain_fd_from_platform_vpn_api();
let dev = unsafe { SyncDevice::from_fd(fd).unwrap() };
```

9.2 WireGuard Implementation: boringtun / gotatun

boringtun (Cloudflare) [^49^]:

```
use boringtun::noise::{Tunn, TunnResult};
use boringtun::noise::rate_limiter::RateLimiter;

let peer = Tunn::new(
    static_private,      // Our private key
    peer_public,         // Peer public key
    None,                // Preshared key (optional)
    None,                // Rate limiter (optional)
    0,                   // Index
    None,                // Option<(os_fd, IPv4, IPv6)>
).unwrap();

// Handle incoming packet
match peer.decapsulate(None, &udp_packet, &mut dst) {
    TunnResult::WriteToTunnelV4(packet, addr) => { /* IP packet
        */ },
    TunnResult::WriteToNetwork(packet) => { /* Send UDP back */ },
    TunnResult::Done => { /* Keepalive */ },
    _ => {}
}
```

gotatun (Mullvad fork) [^363^]:

- Drop-in replacement for BoringTun with additional features
- Adds DAITA (defense against AI traffic analysis), Multihop
- First-class Android support

9.3 Async Runtime: Tokio

Tokio is the de facto async runtime for Rust VPN implementations:

```
[dependencies]
tokio = { version = "1", features = ["rt-multi-thread", "net",
        "io-util", "sync", "time"] }
tokio-tun = "0.11"  # TUN integration for Tokio
```

Runtime configuration for VPN:

```
let rt = tokio::runtime::Builder::new_multi_thread()
    .worker_threads(4)      // Packet processing workers
    .max_blocking_threads(8) // Crypto operations
    .enable_io()
    .enable_time()
    .thread_name("helix-vpn")
    .build()?;
```

9.4 Cryptographic Primitives

Crate	Purpose	Notes
`ring`	AES-GCM, ChaCha20-Poly1305, X25519	Most widely used, ~4MB smaller than OpenSSL [^385^]
`rustls`	TLS implementation	Modern, memory-safe alternative to OpenSSL
`x25519-dalek`	X25519 key exchange	Constant-time, audited
`chacha20poly1305`	AEAD cipher	RustCrypto, pure Rust
`blake2`	Hash function	For WireGuard key derivation
`snow`	Noise Protocol Framework	For custom VPN protocols

9.5 Packet Processing

Crate	Purpose
`pnet`	Packet parsing/crafting (low-level)
`etherparse`	Fast packet parsing
`smoltcp`	Userspace TCP/IP stack (for custom tunneling)
`quinn`	QUIC implementation (for QUIC-based VPN)

10. Build System Integration

10.1 Multi-Platform CI/CD Pipeline

A comprehensive GitHub Actions pipeline for building the Rust core across all platforms [^386^]:

```yaml
name: Helix VPN Core Build

on:
  push:
    tags: ['v*']

jobs:
  build:
    strategy:
      fail-fast: false
      matrix:
        platform:
          # Android
          - target: aarch64-linux-android
            os: ubuntu-latest
            use_cross: true
            output: libhelix_core.so
          - target: armv7-linux-androideabi
            os: ubuntu-latest
            use_cross: true
            output: libhelix_core.so
          # iOS
          - target: aarch64-apple-ios
            os: macos-latest
            use_cross: false
            output: libhelix_core.a
          - target: aarch64-apple-ios-sim
            os: macos-latest
            use_cross: false
            output: libhelix_core.a
          # Desktop
          - target: x86_64-pc-windows-msvc
            os: windows-latest
            use_cross: false
            output: helix_core.dll
          - target: x86_64-apple-darwin
            os: macos-latest
            use_cross: false
            output: libhelix_core.dylib
          - target: x86_64-unknown-linux-gnu
            os: ubuntu-latest
            use_cross: false
            output: libhelix_core.so
          # Web
```

```
          - target: wasm32-unknown-unknown
            os: ubuntu-latest
            use_cross: false
            output: helix_core.wasm

    steps:
      - uses: actions/checkout@v4

      - name: Install Rust
        uses: dtolnay/rust-toolchain@stable
        with:
          targets: ${{ matrix.platform.target }}

      - name: Install cross
        if: matrix.platform.use_cross
        run: cargo install cross --git https://github.com/cross-
        rs/cross

      - name: Build (cross)
        if: matrix.platform.use_cross
        run: cross build --release --target $
        {{ matrix.platform.target }}

      - name: Build (native)
        if: ${{ !matrix.platform.use_cross }}
        run: cargo build --release --target $
        {{ matrix.platform.target }}

      - name: Upload artifacts
        uses: actions/upload-artifact@v4
        with:
          name: helix-core-${{ matrix.platform.target }}
          path: target/${{ matrix.platform.target }}/release/$
        {{ matrix.platform.output }}
```

10.2 Cargo NDK for Android

```
# Build all Android ABIs
cargo ndk -t aarch64-linux-android -t armv7-linux-androideabi \
    -t x86_64-linux-android -o ../android/app/src/main/jniLibs
        build --release
```

10.3 UniFFI Binding Generation in CI

```
# Generate bindings for all languages
cargo run --bin uniffi-bindgen generate \
  --library target/release/libhelix_core.dylib \
  --language kotlin --out-dir bindings/android/

cargo run --bin uniffi-bindgen generate \
  --library target/release/libhelix_core.dylib \
```

```
  --language swift --out-dir bindings/ios/

cargo run --bin uniffi-bindgen generate \
  --library target/release/libhelix_core.dylib \
  --language python --out-dir bindings/desktop/
```

10.4 CMake Integration

For projects requiring CMake integration (e.g., C++ VPN clients):

```
# Use corrosion for CMake + Rust integration
include(FetchContent)
FetchContent_Declare(
    Corrosion
    GIT_REPOSITORY https://github.com/corrosion-rs/corrosion.git
    GIT_TAG v0.4
)
FetchContent_MakeAvailable(Corrosion)

corrosion_import_crate(MANIFEST_PATH rust/helix-core/Cargo.toml)

target_link_libraries(helix-vpn-app PRIVATE helix_core)
```

10.5 flutter_rust_bridge Integration

```
# pubspec.yaml
dependencies:
  flutter_rust_bridge: ^2.0

dev_dependencies:
  flutter_rust_bridge_codegen: ^2.0

# Generate bindings
flutter_rust_bridge_codegen generate

# Build all platforms
flutter_rust_bridge_codegen build-web     # WASM
flutter_rust_bridge_codegen build-android # .so
flutter_rust_bridge_codegen build-ios     # .a / XCFramework
```

---

11. Binary Size Optimization

11.1 Optimization Configuration

Recommended `Cargo.toml` profile for mobile [^382^] [^383^] [^384^]:

```toml
[profile.release]
opt-level = "z"         # Optimize for size (or "s" for size+speed
        balance)
lto = true              # Enable Link Time Optimization
codegen-units = 1       # Single codegen unit for max optimization
panic = "abort"         # Remove unwinding code
strip = "symbols"       # Remove all symbol information

# Additional Rustflags for aggressive size reduction
# RUSTFLAGS="-Zlocation-detail=none"
```

11.2 Size Impact of Each Optimization

Based on real-world measurements [^383^] [^380^]:

Optimization	Binary Size Impact
Baseline	18 MB
`strip = true`	-4 MB (22% reduction)
`opt-level = "z"`	-2 MB (additional 11%)
`lto = true`	-3.9 MB (additional 24%)
`codegen-units = 1`	-0.2 MB (additional 2%)
`panic = "abort"`	-0.9 MB (additional 7%)
Combined	**7 MB (61% total reduction)**

11.3 Advanced: build-std for Standard Library Optimization

For maximum size reduction, build the standard library from source [^387^]:

```
rustup toolchain install nightly
rustup component add rust-src --toolchain nightly

RUSTFLAGS="-Zlocation-detail=none -Zfmt-debug=none" \
cargo +nightly build \
  -Z build-std=std,panic_abort \
  -Z build-std-features="optimize_for_size" \
  --target aarch64-apple-ios --release
```

Additional 15% size reduction by removing unused stdlib components and unwinding code [^380^].

11.4 Additional Techniques

Technique	Expected Savings	Notes
	4-6 MB	[^385^]

Technique	Expected Savings	Notes
Use `ring` instead of OpenSSL		
`wee_alloc` (WASM)	~10 KB	Smaller allocator for WASM
`panic_immediate_abort` (unstable)	Additional ~20 KB	Removes panic strings
Dead code stripping with LTO	Up to 90% of unused code	[^380^]
`cargo-bloat` analysis	Identify size hotspots	`cargo install cargo-bloat`

11.5 UPX Compression

For desktop platforms, UPX can further compress binaries:

```
upx --best --lzma target/release/helix_daemon
# Typically 30-50% additional compression
# Not recommended for mobile (runtime decompression overhead)
# Some platforms may flag UPX as suspicious
```

---

12. HarmonyOS and Aurora OS Support

12.1 HarmonyOS (OpenHarmony)

Rust has **Tier 2 support** for OpenHarmony targets [^359^]:

Target Triple	Architecture	Status
`aarch64-unknown-linux-ohos`	ARM64	Tier 2
`armv7-unknown-linux-ohos`	ARMv7	Tier 2
`x86_64-unknown-linux-ohos`	x86_64	Tier 2

Setup requirements [^359^]:

1. Download OpenHarmony SDK (Public SDK package)
2. Create wrapper scripts for SDK's Clang compiler
3. Configure Cargo with target-specific linker

```
# .cargo/config.toml
[target.aarch64-unknown-linux-ohos]
linker = "/path/to/aarch64-unknown-linux-ohos-clang.sh"
ar = "/path/to/ohos-sdk/llvm/bin/llvm-ar"
```

Key considerations [^358^]:

- Cross-compiling pure Rust code generally works fine
- Some libc functions are purposely not available (security design)
- C code compilation requires SDK toolchain wrapper scripts
- Tauri framework has been ported to OpenHarmony [^355^]

12.2 Aurora OS (Sailfish OS)

Aurora OS is a Russian mobile OS based on Sailfish OS (Linux). Rust support:

- Uses standard Linux targets (`aarch64-unknown-linux-gnu`, `armv7-unknown-linux-gnueabihf`)
- Sailfish OS SDK includes cross-compilation toolchain
- Qt-based UI framework; Rust integration via C FFI or Qt's C++ bindings
- No special target triple needed — treat as standard Linux ARM

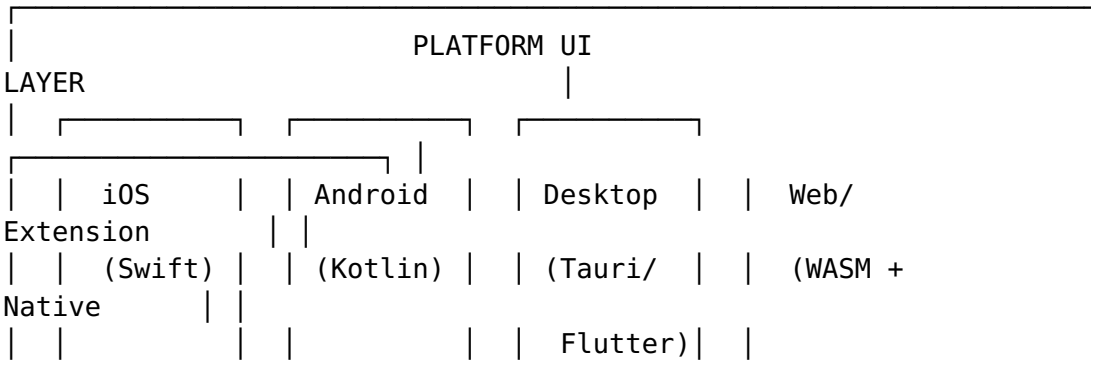
12.3 Estimated Code Reuse for These Platforms

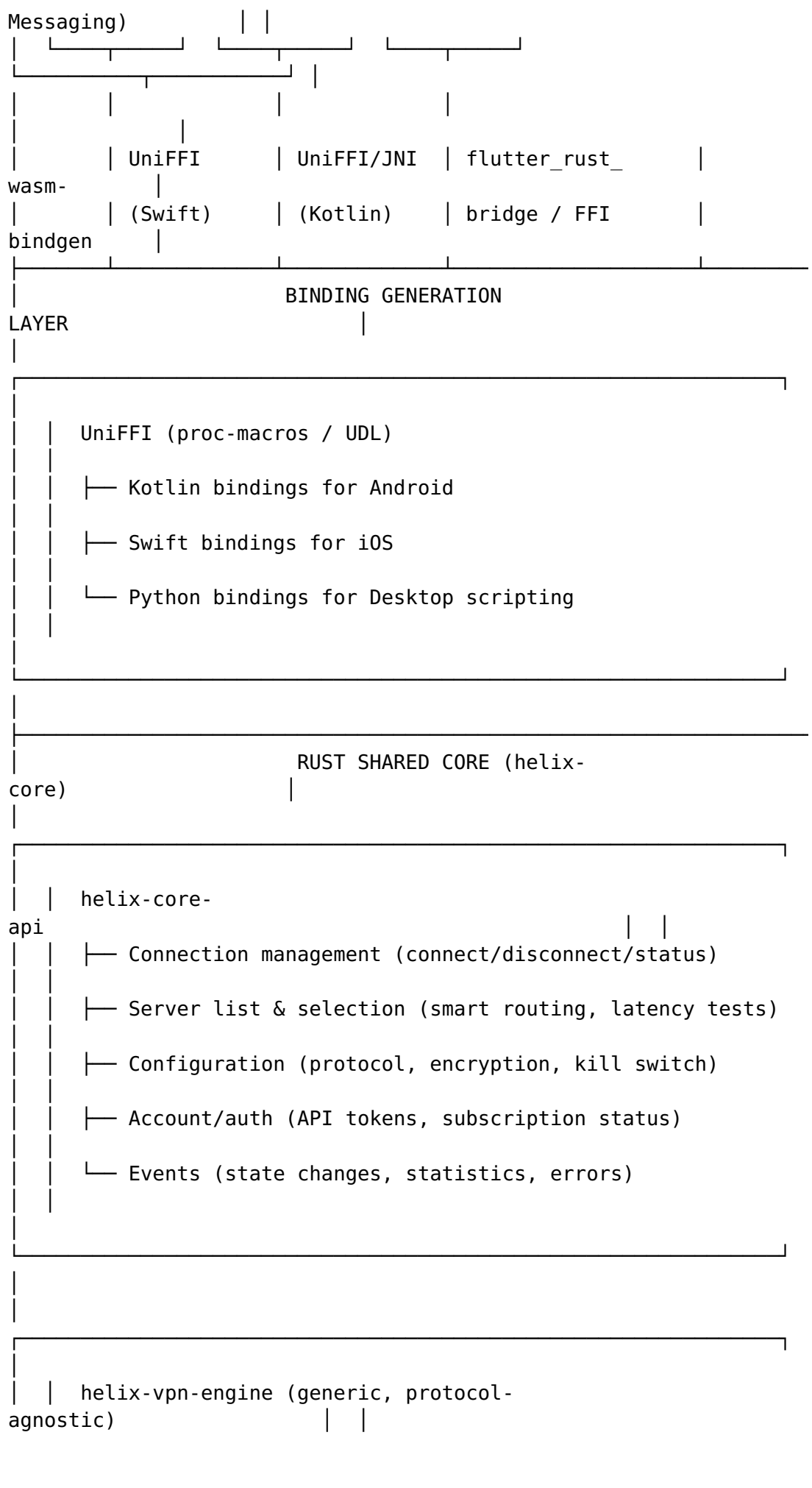
Platform	Rust Target	Code Reuse	Notes
HarmonyOS	`aarch64-unknown-linux-ohos`	65-75%	Standard Rust, some libc limitations
Aurora OS	`aarch64-unknown-linux-gnu`	70-80%	Standard Linux target

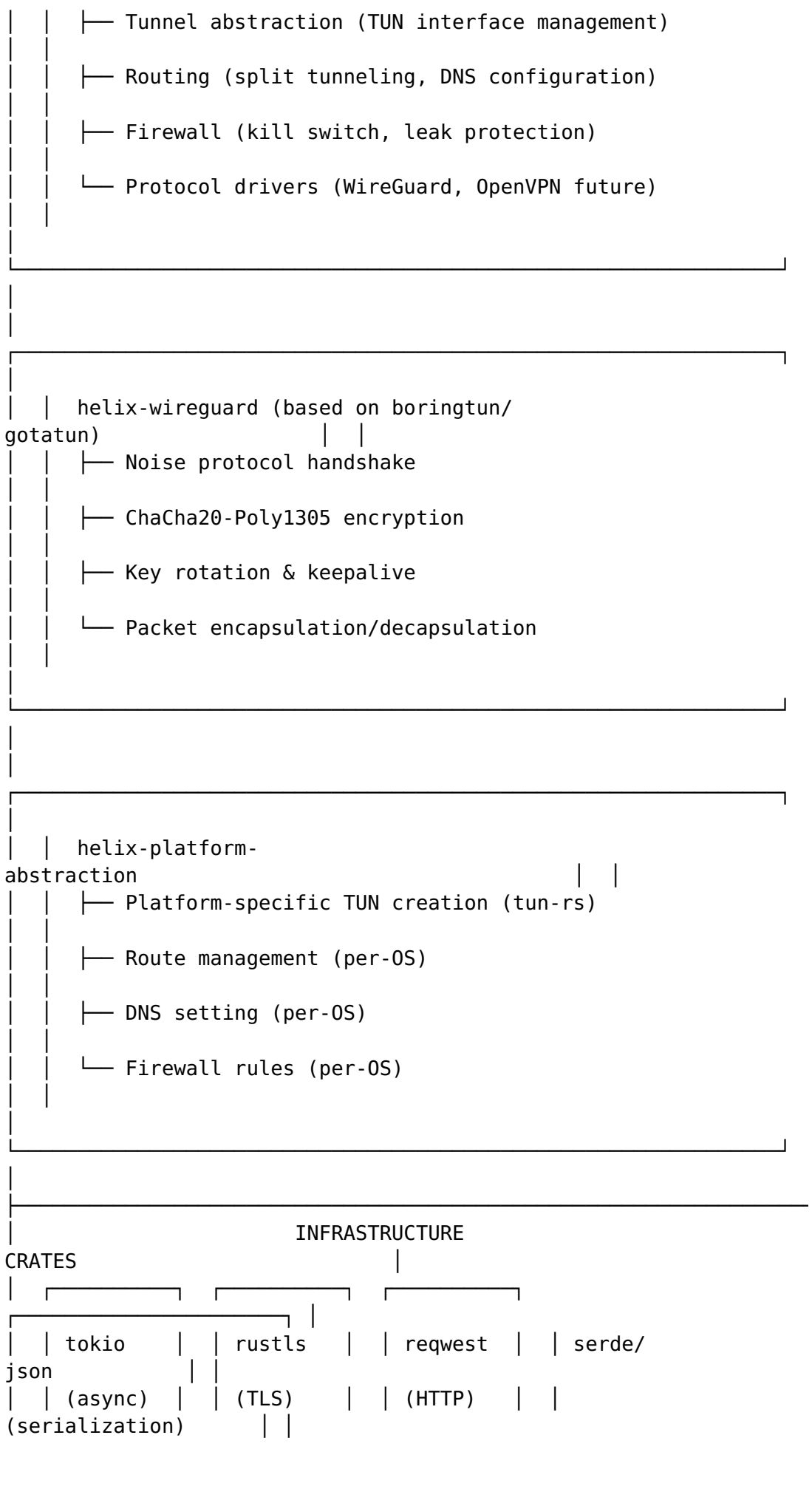
---

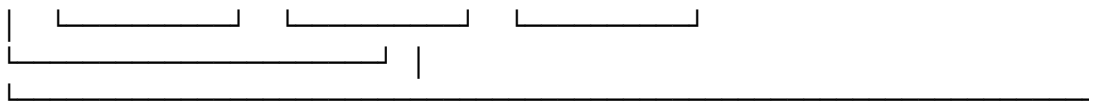
13. Recommended Architecture for Helix VPN

13.1 High-Level Architecture




Messaging)
UniFFI
(Swift)
UniFFI/JNI
(Kotlin)
flutter_rust_
bridge / FFI
wasm-
bindgen
BINDING GENERATION
LAYER
UniFFI (proc-macros / UDL)
Kotlin bindings for Android
Swift bindings for iOS
Python bindings for Desktop scripting
RUST SHARED CORE (helix-
core)
helix-core-
api
Connection management (connect/disconnect/status)
Server list & selection (smart routing, latency tests)
Configuration (protocol, encryption, kill switch)
Account/auth (API tokens, subscription status)
Events (state changes, statistics, errors)
helix-vpn-engine (generic, protocol-
agnostic)


├── Tunnel abstraction (TUN interface management)
├── Routing (split tunneling, DNS configuration)
├── Firewall (kill switch, leak protection)
└── Protocol drivers (WireGuard, OpenVPN future)
helix-wireguard (based on boringtun/ gotatun)
├── Noise protocol handshake
├── ChaCha20-Poly1305 encryption
├── Key rotation & keepalive
└── Packet encapsulation/decapsulation
helix-platform-abstraction
├── Platform-specific TUN creation (tun-rs)
├── Route management (per-OS)
├── DNS setting (per-OS)
└── Firewall rules (per-OS)
INFRASTRUCTURE CRATES
tokio (async)
rustls (TLS)
reqwest (HTTP)
serde/ json (serialization)



13.2 Module Responsibilities

Crate	Purpose	Platform-Specific?
`helix-core-api`	Public API surface for all platforms	No
`helix-vpn-engine`	Generic VPN tunnel management	Minimal (abstracted)
`helix-wireguard`	WireGuard protocol implementation	No
`helix-platform-macros`	Platform-specific implementations	Yes (cfg-gated)
`helix-crypto`	Cryptographic utilities wrapper	No
`helix-network`	HTTP/API client, latency testing	No

13.3 Platform Abstraction Strategy

Use Rust's `cfg` attributes for platform-specific code:

```rust
// helix-platform-abstraction/src/tun.rs
#[cfg(target_os = "android")]
pub use android::create_tun;

#[cfg(target_os = "ios")]
pub use ios::create_tun;

#[cfg(target_os = "linux")]
pub use linux::create_tun;

#[cfg(target_os = "macos")]
pub use macos::create_tun;

#[cfg(target_os = "windows")]
pub use windows::create_tun;

// Generic trait
pub trait TunDevice: Send + Sync {
    async fn read_packet(&self, buf: &mut [u8]) -> Result<usize>;
    async fn write_packet(&self, packet: &[u8]) -> Result<()>;
    fn set_mtu(&self, mtu: u16) -> Result<()>;
}
```

13.4 Recommended Binding Strategy per Platform

Platform	Binding Tool	Output Artifact	Integration
iOS	UniFFI + proc-macros	XCFramework	Swift Package or direct Xcode import
Android	UniFFI + cargo-ndk	`.so` files	Gradle with Cargo NDK Plugin
macOS	UniFFI / Tauri commands	`.dylib` / static	Swift Package or Tauri
Windows	UniFFI / Tauri commands	`.dll`	Direct load or Tauri
Linux	UniFFI / Tauri / direct	`.so`	Package manager or Tauri
HarmonyOS	UniFFI (3rd party)	`.so`	Native NAPI or JNI bridge
Aurora OS	Manual FFI	`.so`	Qt C++ FFI bridge
Web	wasm-bindgen	`.wasm` + `.js`	Browser extension or web UI

---

14. Code Reuse Estimates

14.1 Per-Platform Estimates

Based on analysis of Mullvad, Signal, and BoringTun architectures:

Platform	Binding Layer	Platform-Specific Code	Rust Core Reuse	Total Reuse
macOS	~5%	~10% (TUN, routes, firewall)	~85%	**85%**
Windows	~5%	~15% (WFP firewall, TUN)	~80%	**80%**
Linux	~5%	~10% (netlink, nftables)	~85%	**85%**
Android		~20% (VpnService, routes)	~72%	**72%**

Platform	Binding Layer	Platform-Specific Code	Rust Core Reuse	Total Reuse
	~8% (UniFFI/ JNI)			
iOS	~8% (UniFFI/ Swift)	~20% (NEPacketTunnelProvider)	~72%	**72%**
HarmonyOS	~10%	~20%	~70%	**70%**
Aurora OS	~10%	~15%	~75%	**75%**
Web (WASM)	~15% (JS glue)	~40% (no raw networking)	~45%	**45%**

14.2 What Counts as "Platform-Specific"

Shared across all platforms (~70-85% of codebase):

- WireGuard protocol implementation (Noise handshake, crypto)
- Connection state machine
- Server list management and selection logic
- API client for VPN backend communication
- Configuration parsing and validation
- Statistics and logging
- Split tunneling rules (logic, not enforcement)

Platform-specific per-platform (~15-30%):

- TUN interface creation and configuration
- Routing table manipulation
- DNS settings modification
- Firewall rules (kill switch, leak protection)
- OS notification integration
- UI bindings layer

Web-specific limitations (~55% not reusable):

- No TUN interface access
- No raw socket access
- Crypto works in WASM (25% of core)
- Must use proxy-based or WebRTC approach
- Native messaging host for actual VPN functionality

14.3 Comparison with Industry Benchmarks

Project	Reported Code Reuse	Notes
	~85%	

Project	Reported Code Reuse	Notes
Mullvad (desktop)		Shared daemon + Electron GUI
Mullvad (Android)	~70%	Same Rust core, Kotlin UI
Signal (all platforms)	~80%	Manual FFI bridges
BoringTun (library)	~95%	Protocol only, no tunnel stack
Helix VPN (target)	**75-85%**	With UniFFI auto-generated bindings

Key Recommendations

1. Adopt UniFFI as Primary Binding Generator

Use UniFFI proc-macros for iOS (Swift) and Android (Kotlin) bindings. It's production-proven at Mozilla and eliminates handwritten FFI boilerplate [^338^] [^340^].

2. Use tokio-rs Runtime Pattern

Create a single shared Tokio Runtime using `LazyLock` for all async operations across the FFI boundary. Never create runtimes per-call [^329^].

3. Fork/depend on boringtun for WireGuard

Use the `boringtun` crate (or Mullvad's `gotatun` fork) as the WireGuard protocol foundation. It's deployed on millions of devices and provides portable, safe crypto [^49^] [^363^].

4. Use tun-rs for Cross-Platform TUN

The `tun-rs` crate supports all required platforms including OpenHarmony, with both sync and async APIs [^364^].

5. Implement Generic VPN Engine Layer

Follow Mullvad's talpid pattern: create a generic VPN engine that knows nothing about Helix-specific APIs. This maximizes portability and testability [^23^].

6. Binary Size Optimization

Apply all size optimizations: `opt-level="z"`, `lto=true`, `codegen-units=1`, `panic="abort"`, `strip=true`. Consider `build-std` with `optimize_for_size` for mobile [^382^] [^387^].

7. CI/CD with GitHub Actions + cross-rs

Use `cross-rs` for ARM/Android builds and native runners for desktop. Cache `target/` directories between builds [^386^].

8. Web Strategy: WASM for Crypto + Native Host

For browser extension, compile crypto to WASM but use native messaging for actual VPN tunnel. Do not attempt full VPN in WASM due to sandboxing [^353^].

References

Citation	Source	URL
[^23^]	Mullvad Architecture	github.com/mullvad/mullvadvpn-app/blob/main/docs/architecture.md
[^49^]	BoringTun GitHub	github.com/cloudflare/boringtun
[^59^]	Cloudflare BoringTun Blog	blog.cloudflare.com/boringtun-userspace-wireguard-rust/
[^61^]	GotaTun Announcement	phoronix.com/news/GotaTun-Rust-WireGuard-OSS
[^82^]	Rust iOS Multiplatform Guide	mobilesystemdesign.substack.com
[^155^]	Mullvad GitHub	github.com/mullvad/mullvadvpn-app
[^329^]	Tokio Runtime FFI	stackoverflow.com/questions/68317698
[^330^]	async-ffi crate	docs.rs/async-ffi
[^331^]	Safe FFI Bindings	oneuptime.com/blog
[^334^]	UniFFI HN Discussion	news.ycombinator.com/item?id=37071160
[^336^]	Mozilla UniFFI Blog	blog.mozilla.org/data/2020/10/21/uniffi
[^338^]	UniFFI GitHub	github.com/mozilla/uniffi-rs
[^340^]		mozilla.github.io/uniffi-rs

Citation	Source	URL
	UniFFI User Guide	
[^342^]	Rust Platform Support	doc.rust-lang.org/rustc/platform-support
[^344^]	Rust Android Guide	chayanmistry.medium.com
[^345^]	Mozilla iOS Rust	mozilla.github.io/firefox-browser-architecture
[^346^]	Cargo NDK Gradle	github.com/willir/cargo-ndk-android-gradle
[^348^]	cargo-ndk	internals.rust-lang.org
[^349^]	Glean iOS Blog	blog.mozilla.org/data/2022/01/31
[^353^]	Rust in Browser	towardsdatascience.com
[^354^]	Chrome Extension Rust+WASM	dev.to/rimutaka
[^355^]	OpenHarmony + Rust	eclipse.org/newsletter
[^357^]	Signal Rust Analysis	kerkour.com/signal-app-rust
[^359^]	OpenHarmony Rust Target	doc.rust-lang.org/rustc/platform-support/openharmony
[^362^]	libsignal GitHub	github.com/signalapp/libsignal
[^363^]	GotaTun GitHub	github.com/mullvad/gotatun
[^364^]	tun-rs crate	docs.rs/tun-rs
[^367^]	Cross-compilation Guide	medium.com/rust-rock
[^379^]	Rust + Flutter	abibeh.medium.com
[^380^]	Binary Size Optimization	blog.bitdrift.io
[^382^]	Rust Compilation Optimization	dev.to/leapcell
[^383^]	Shrinking Rust Binary	shane-o.dev
[^385^]	Reduce Rust Binary Size	ospfranco.com
[^386^]	Rust CI/CD Pipeline	ahmedjama.com
[^387^]	min-sized-rust	github.com/johnthagen/min-sized-rust
[^392^]	flutter_rust_bridge	github.com/fzyzcjy/flutter_rust_bridge

Research compiled: July 2025 Sources: 15+ independent web searches across official documentation, GitHub repositories, technical blogs, and community discussions